Hankuk University
CHEM 101 General Chemistry I
Credits: 3

Course Description
Fundamental principles of chemistry: atoms, stoichiometry, chemical bonding, gases and thermochemistry.

Learning Outcomes
1. Perform stoichiometric calculations using the mole concept.
2. Describe electron configurations and periodic trends.
3. Draw Lewis structures and predict molecular geometry using VSEPR.
4. Apply the ideal gas law and calculate enthalpy changes.

Textbook
Brown, LeMay and Bursten, Chemistry: The Central Science, 15th edition

Weekly Schedule
Week 1: Matter, measurement and significant figures
Week 2: Atoms, molecules and ions
Week 3: Chemical equations and stoichiometry
Week 4: Reactions in aqueous solution
Week 5: Thermochemistry: enthalpy and calorimetry
Week 6: Electronic structure of atoms
Week 7: Periodic properties of the elements
Week 8: Midterm exam
Week 9: Chemical bonding: ionic and covalent bonds, Lewis structures
Week 10: Molecular geometry and bonding theories, VSEPR
Week 11: Gases: the ideal gas equation
Week 12: Intermolecular forces
Week 13: Properties of solutions
Week 14: Review
Week 15: Final exam

Assessment
Quizzes 15%, Midterm 35%, Final exam 50%